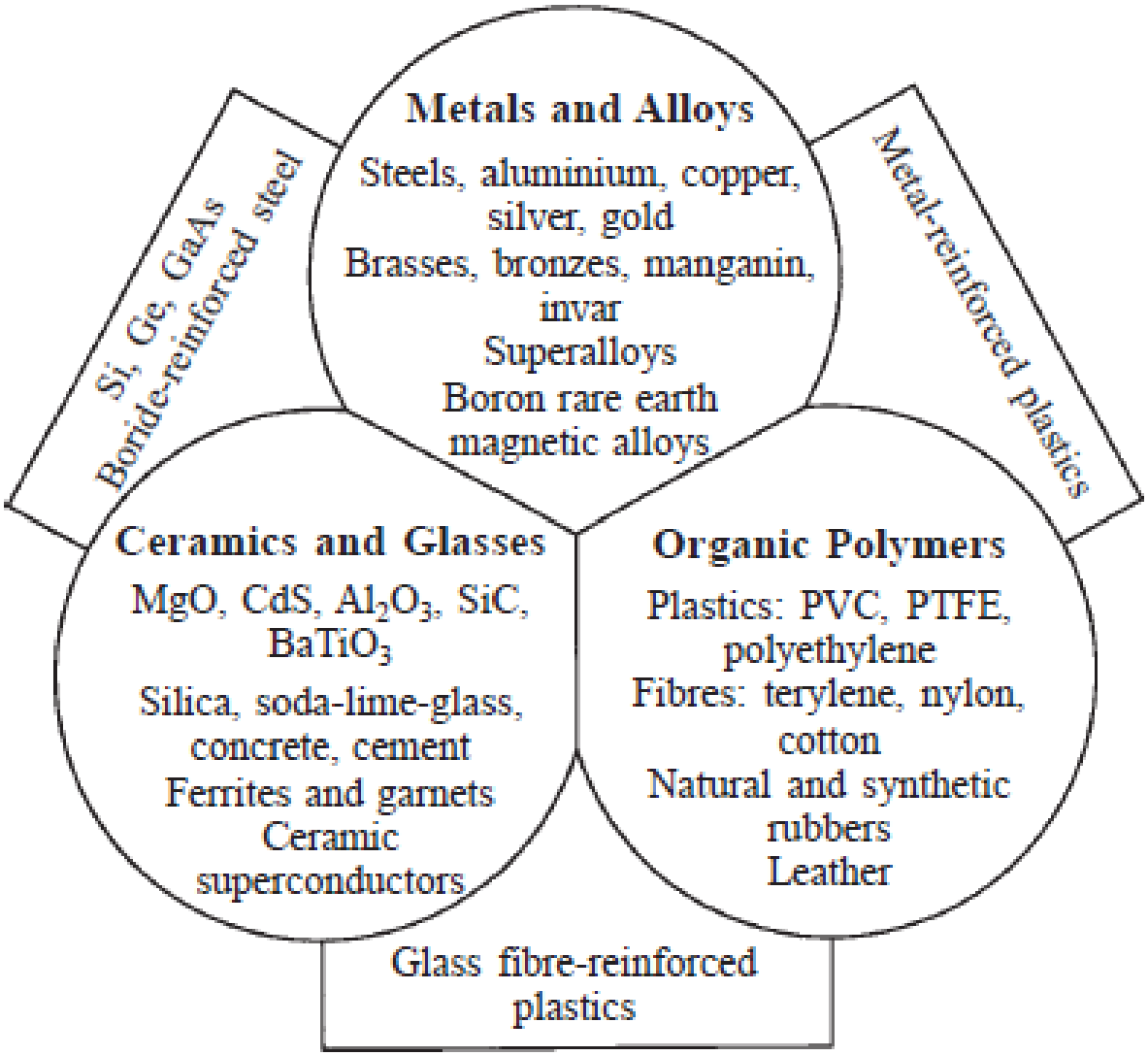
Introduction and Structures of Materials - I



MO-201 Materials Science
Odd Sem 2026-27

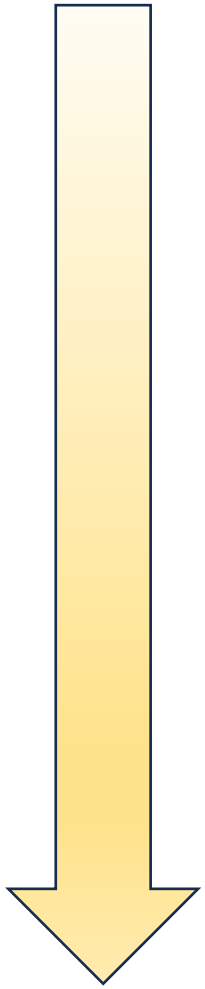
Dr. S. Sinha, Dept. of Metallurgical Engg.
IIT (BHU) Varanasi

Classification of Engineering Materials



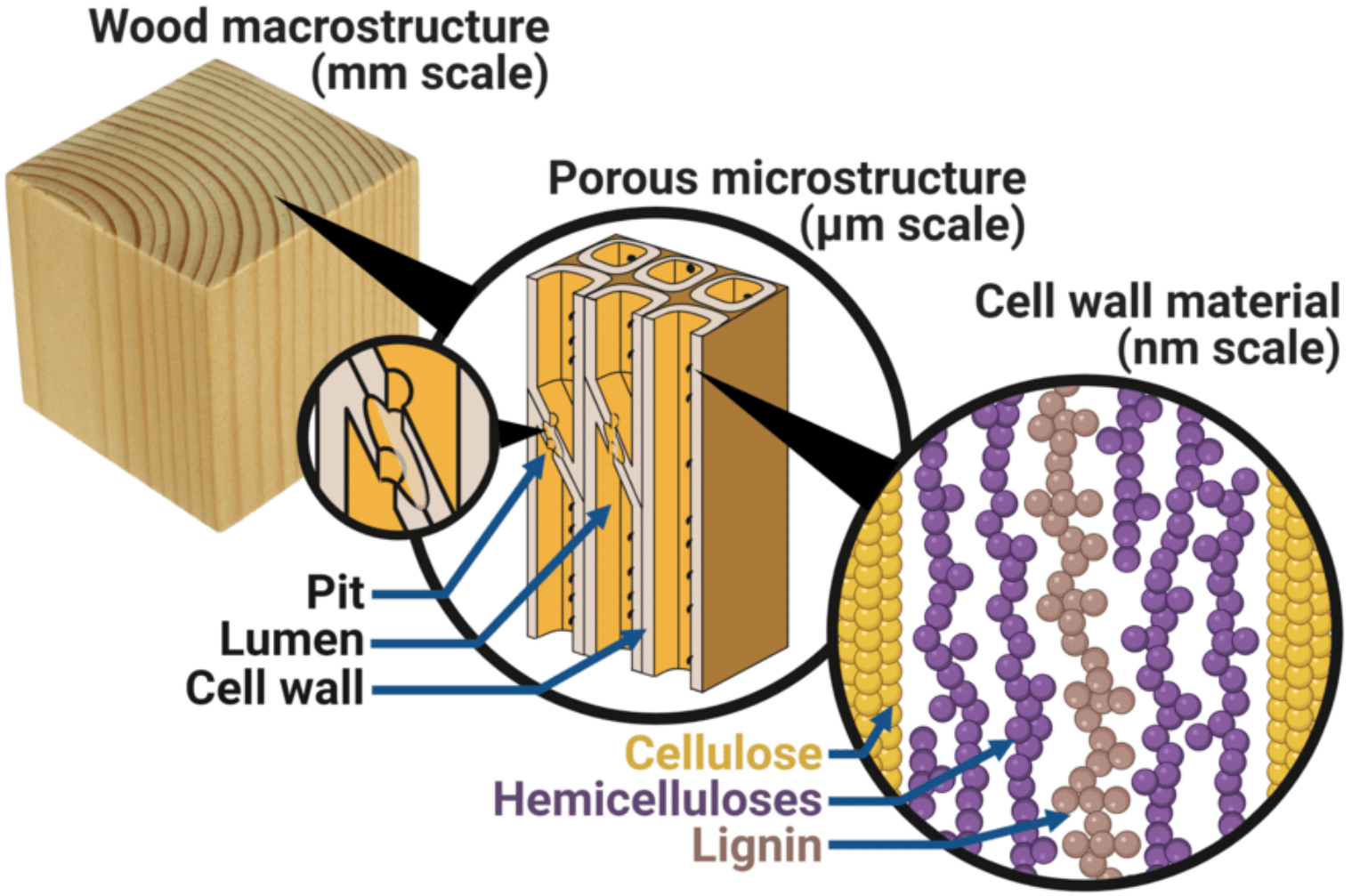
Levels of Structure

- Macrostructure
- Microstructure
- Substructure
- Crystal structure
- Electronic structure
- Nuclear structure



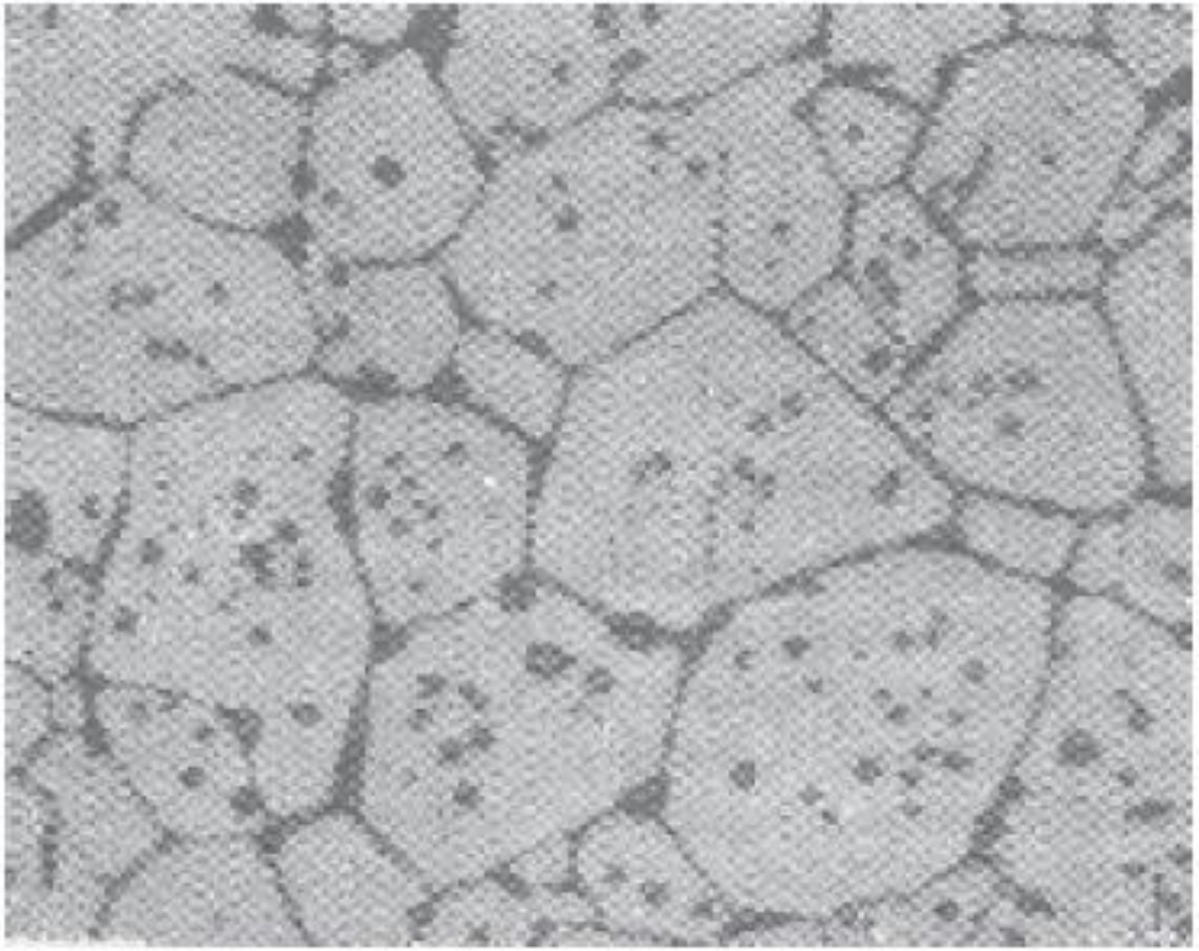
Decreasing length scale

Levels of Structure



Thybring & Fredriksson, Wood Modification as a Tool to Understand Moisture in Wood, Forests 12 (2021) 372

Levels of Structure



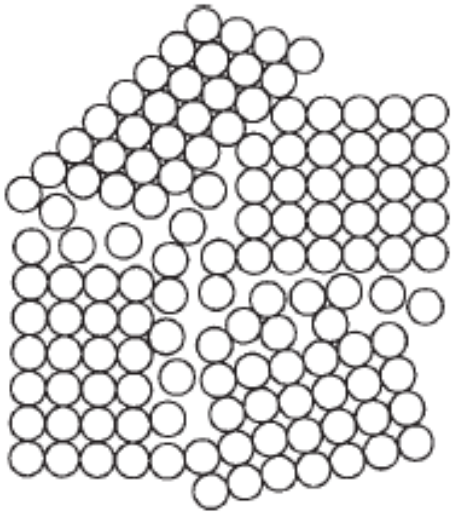
Crystal boundaries in nickel ferrite, Fe_2NiO_4, magnified 900 times linear

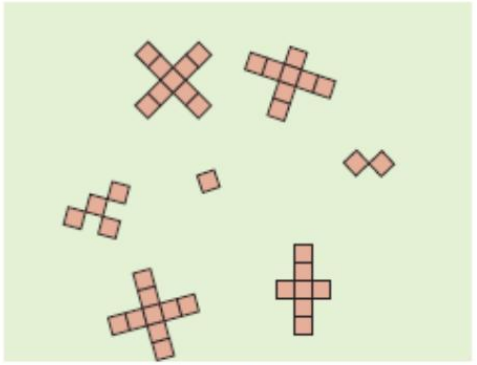
Levels of Structure



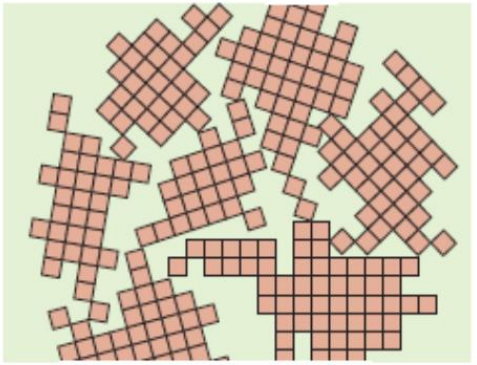
Substructure of a Ni–Fe–Cr alloy showing curved dislocation lines, magnified 30 000 times in an electron microscope

Levels of Structure

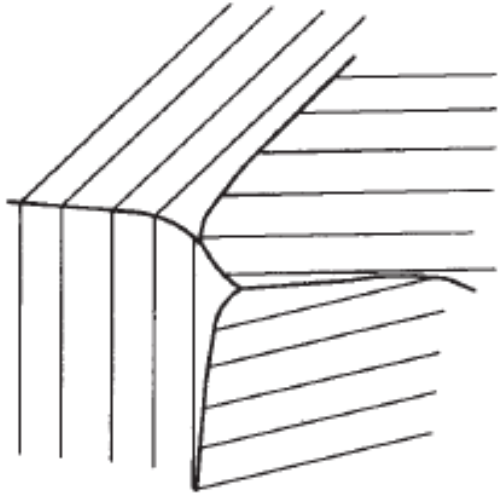


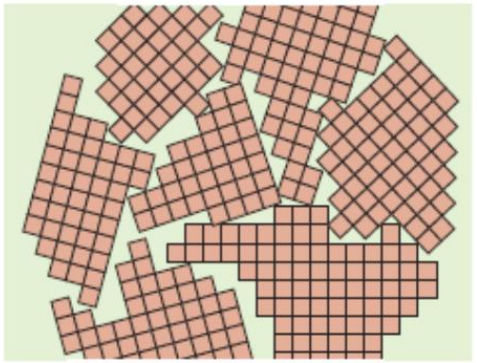


(a)

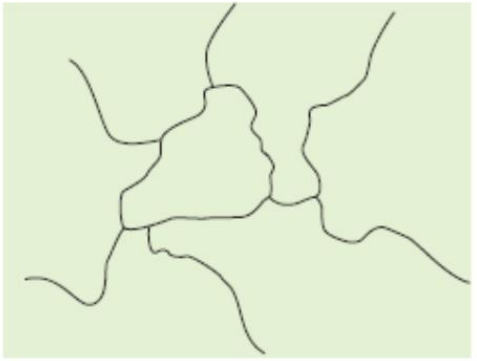


(b)





(c)

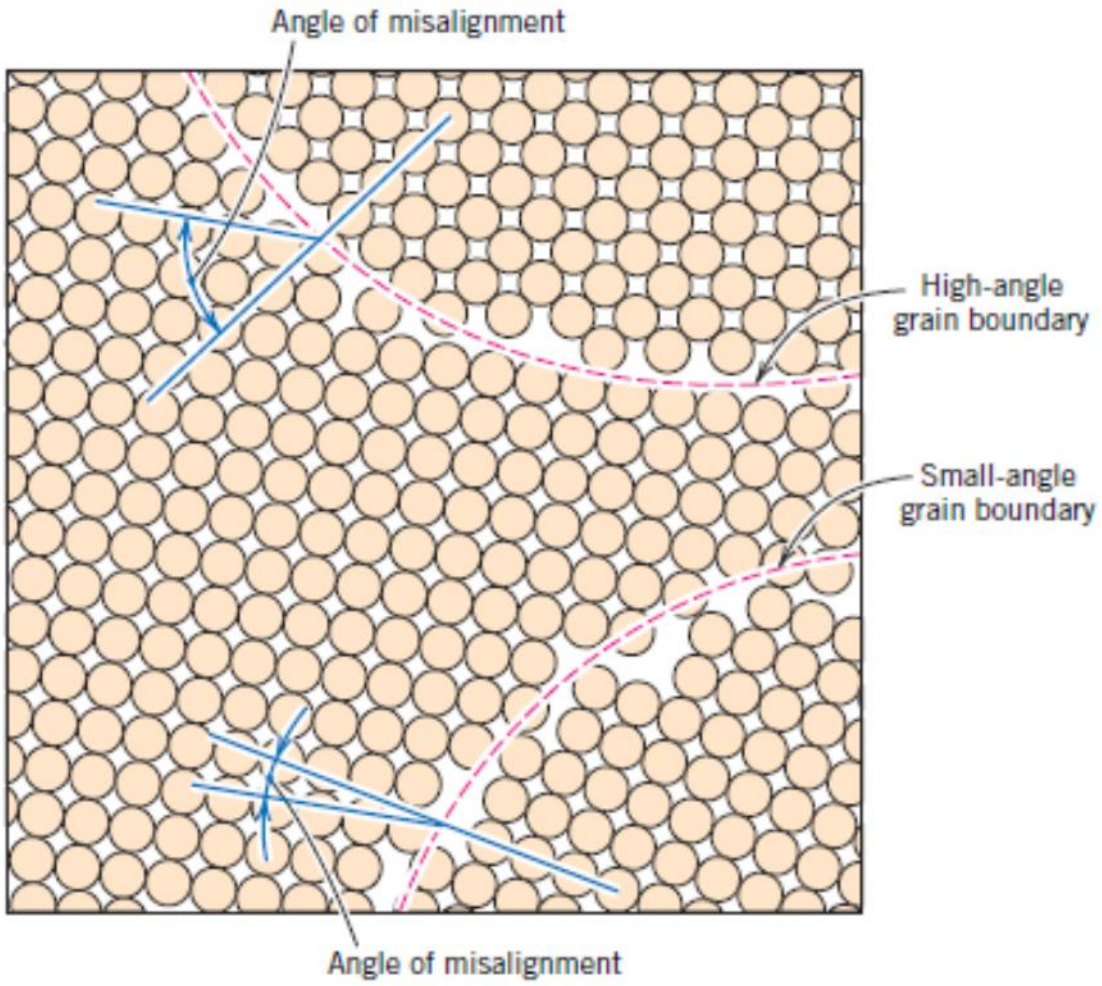


(d)

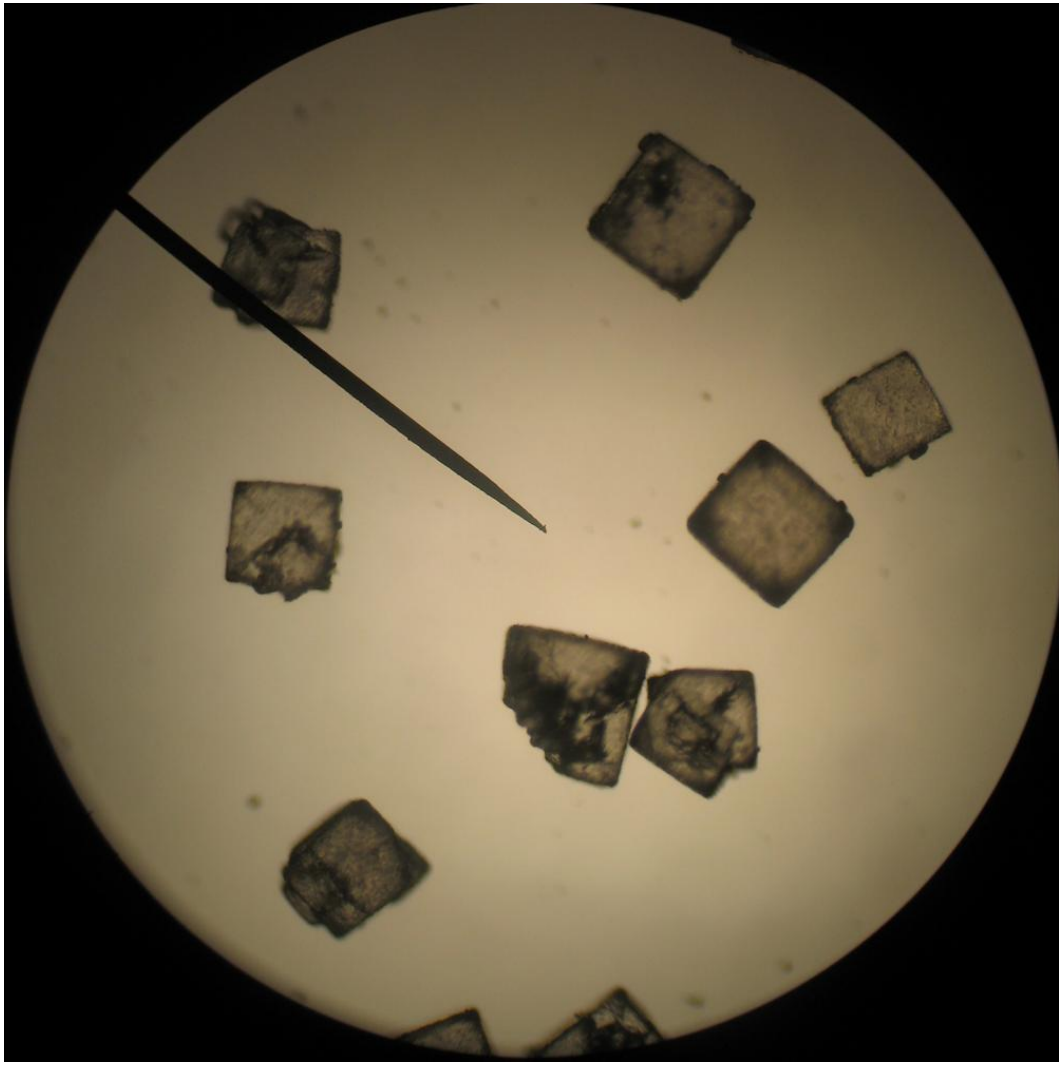
Schematic showing atomic arrangements at grain boundaries

Levels of Structure

Grain boundaries



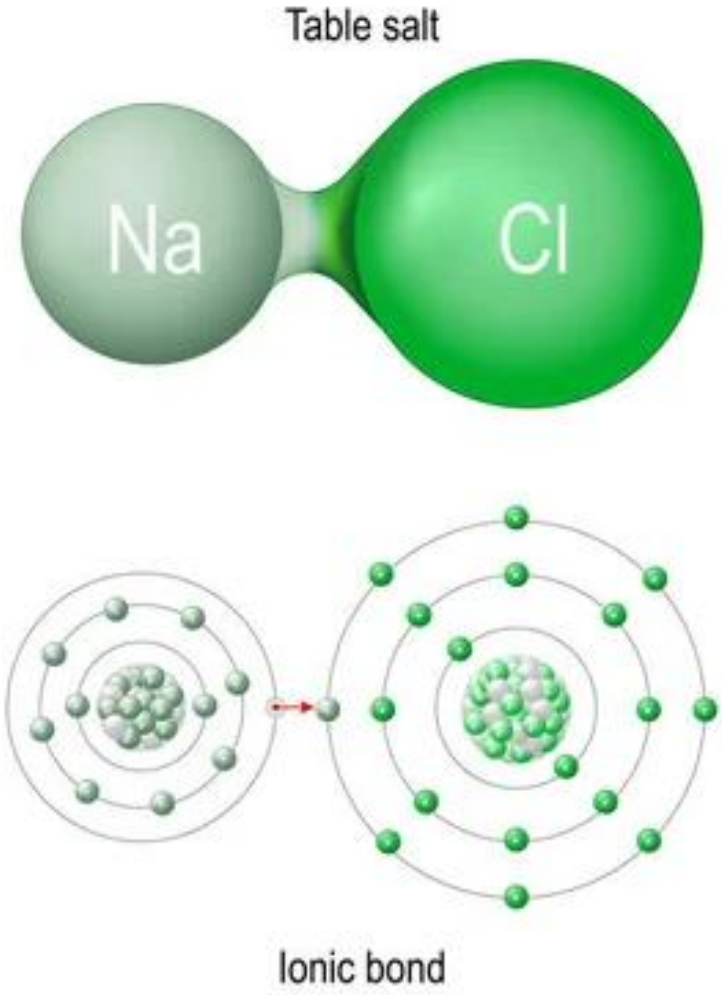
Levels of Structure

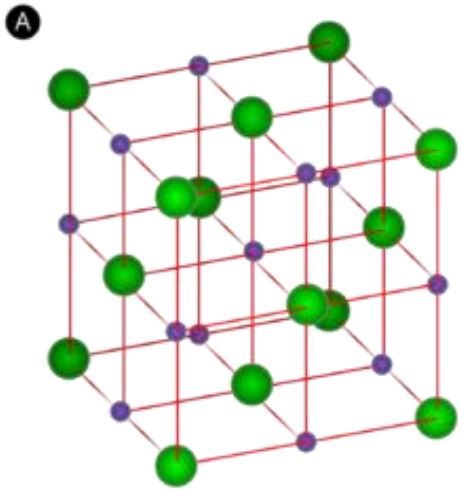


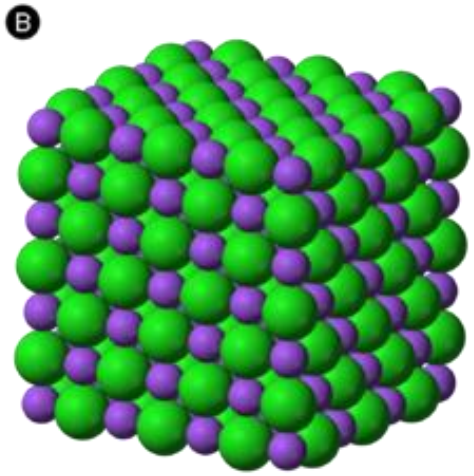
Cube shaped salt crystals under 40x magnification

Picture Courtesy: Salt and Sugar Under the Microscope, Lensyl Urban, Natural World, 2011

Levels of Structure

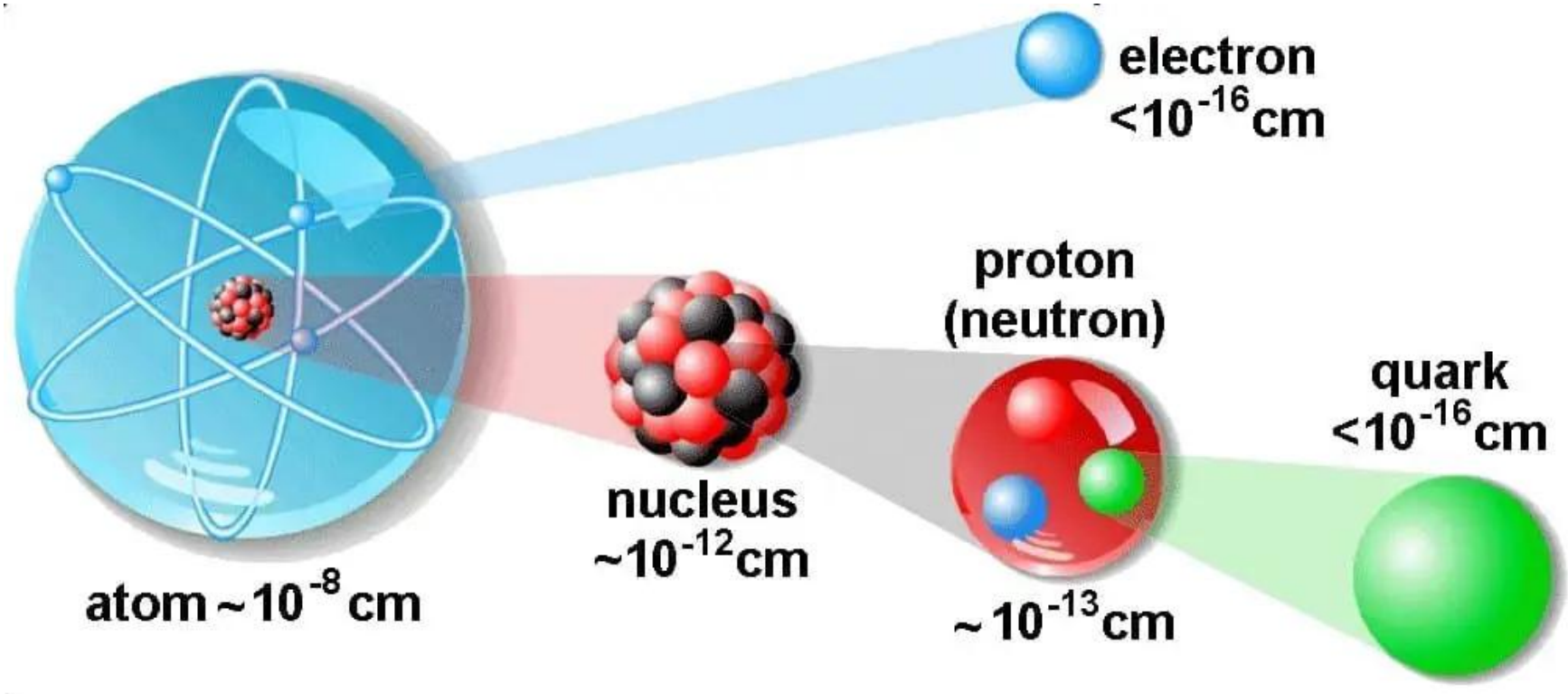






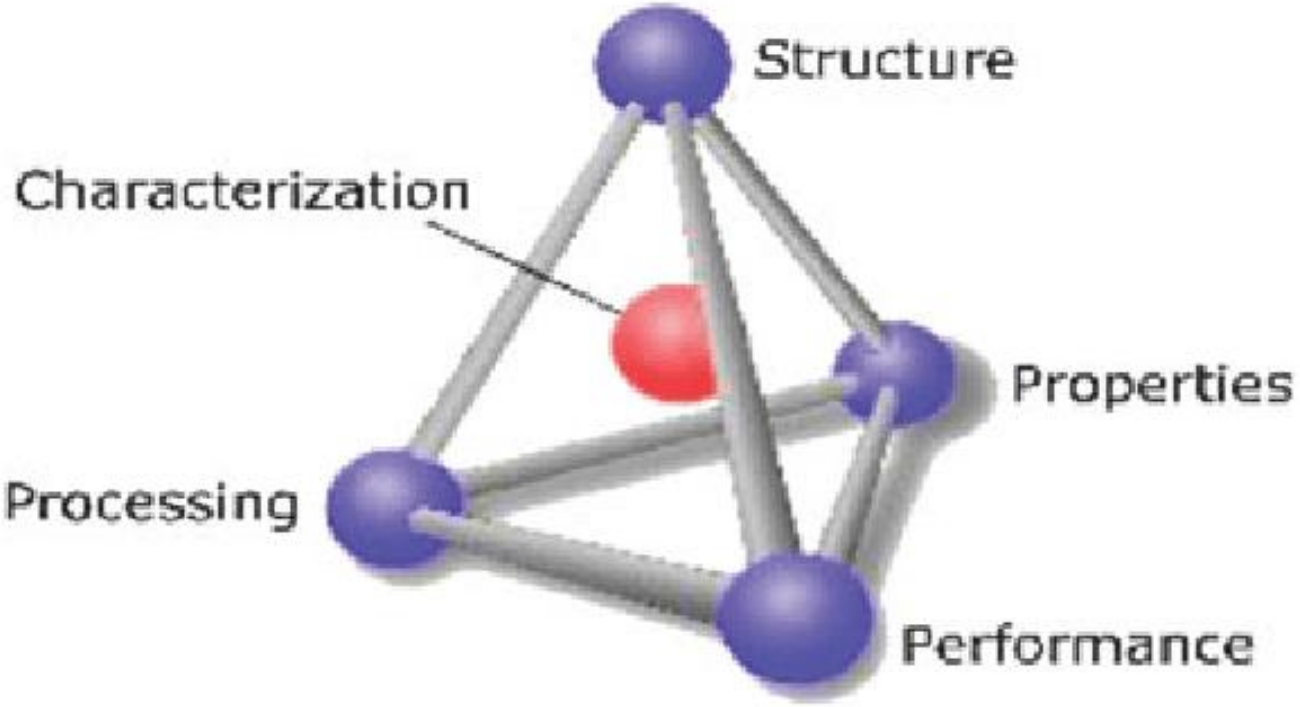
Pictures courtesy:
https://www.istockphoto.com/vector/ionic-vs-covalent-bonds-gm1301181932-393288502
https://www.coursehero.com/study-guides/cheminter/cation-formation-2/

Levels of Structure



https://www.nuclear-power.com/nuclear-power/reactor-physics/atomic-nuclear-physics/atomic-nuclear-structure/

Traditional Materials Science Tetrahedron



J.O. Cross et al., Materials characterization and the evolution of materials, MRS Bulletin 40 (2015) 1019

References

- Materials Science and Engineering, A First Course by V. Raghavan

- Fundamentals of Materials Science and Engineering, W.D. Callister